Results for XES3G5M-sub-small

theta_m=0.8, theta_f=0.8, rec=50 | N=5,10,...,50 | Ep thresholds 0.4, 0.5, 0.6, 0.7

Each table ranks models within the same graph and student split, using unrounded values.
Bold = best; underline = second best. Ties at the best value are all bold.

Experimental groups

Group	Models	exfr train edges	Test students
1. Complete forgetting graph	8/8	1,167,562	241
2. Remove zero-forgetting edges	8/8	694,068	241
3. Weighted nonzero sampling, up to 400/student	8/8	290,800	241
4. Weighted nonzero sampling, up to 200/student	8/8	145,400	241

Model columns

2CKG4ER, No-mastery, No-forgetting, No-student-feature, No-exercise-feature, ID-based only

TransE and TransE-adv appear only in groups with eight completed models.

Source locations

Data_Fin_New/_forgetting_edge_runs_rec50/full_graph/ + _forgetting_edge_comparison_runs_rec50/full_graph/

Data_Fin_New/_forgetting_edge_runs_rec50/drop_exfr0p00/ + _forgetting_edge_comparison_runs_rec50/drop_exfr0p00/

Data_Fin_New/_forgetting_edge_runs_rec50/weighted_nonzero_400/ + _forgetting_edge_comparison_runs_rec50/weighted_nonzero_400/

Data_Fin_New/_forgetting_edge_runs_rec50/weighted_nonzero_200/ + _forgetting_edge_comparison_runs_rec50/weighted_nonzero_200/

Group 1: Complete forgetting graph

Train edges: mastery 279,895; forgetting 1,167,562; recommendation 36,350 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	**0.877**	0.801	0.868	0.856	0.875	0.852	0.843	0.845
10	**0.876**	0.805	0.876	0.860	0.874	0.852	0.846	0.846
15	**0.874**	0.816	0.873	0.859	0.872	0.852	0.847	0.846
20	0.870	0.824	**0.874**	0.858	0.870	0.853	0.847	0.844
25	0.869	0.829	**0.872**	0.857	0.868	0.852	0.845	0.843
30	0.867	0.833	**0.872**	0.856	0.868	0.853	0.844	0.843
35	0.867	0.837	**0.872**	0.854	0.866	0.854	0.844	0.842
40	0.867	0.840	**0.873**	0.853	0.866	0.854	0.843	0.842
45	0.866	0.843	**0.873**	0.852	0.866	0.854	0.843	0.842
50	0.866	0.844	**0.874**	0.852	0.865	0.854	0.843	0.841

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	1.000	**1.000**	0.996	**1.000**	**1.000**	0.999	**1.000**	**1.000**
10	**1.000**	1.000	0.998	1.000	**1.000**	0.999	**1.000**	**1.000**
15	**1.000**	1.000	0.996	1.000	**1.000**	0.999	**1.000**	**1.000**
20	**1.000**	1.000	0.996	1.000	1.000	0.999	**1.000**	**1.000**
25	**1.000**	1.000	0.996	1.000	1.000	1.000	**1.000**	**1.000**
30	**1.000**	1.000	0.995	1.000	1.000	1.000	**1.000**	**1.000**
35	1.000	1.000	0.995	1.000	1.000	1.000	**1.000**	**1.000**
40	1.000	1.000	0.994	1.000	1.000	1.000	**1.000**	**1.000**
45	1.000	1.000	0.994	1.000	1.000	1.000	**1.000**	**1.000**
50	1.000	1.000	0.993	1.000	1.000	1.000	**1.000**	**1.000**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 1: Complete forgetting graph

Train edges: mastery 279,895; forgetting 1,167,562; recommendation 36,350 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.180	**0.201**	-0.001	0.161	0.112	0.157	0.199	0.170
10	0.228	**0.327**	0.002	0.254	0.249	0.255	0.317	0.264
15	0.345	**0.472**	0.063	0.296	0.342	0.313	0.371	0.329
20	0.397	**0.467**	0.029	0.333	0.353	0.341	0.411	0.380
25	0.441	**0.465**	0.050	0.357	0.372	0.373	0.386	0.376
30	0.448	**0.492**	0.140	0.361	0.392	0.377	0.391	0.376
35	0.450	**0.459**	0.201	0.377	0.390	0.387	0.397	0.372
40	**0.444**	0.408	0.260	0.376	0.385	0.398	0.394	0.376
45	**0.452**	0.439	0.331	0.379	0.388	0.397	0.397	0.374
50	**0.421**	0.413	0.337	0.381	0.405	0.400	0.391	0.370

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.178	0.195	-0.004	0.160	0.109	0.151	**0.195**	0.164
10	0.227	**0.322**	-0.010	0.253	0.246	0.251	0.313	0.258
15	0.344	**0.468**	0.049	0.294	0.340	0.308	0.366	0.324
20	0.397	**0.463**	0.013	0.330	0.351	0.336	0.407	0.375
25	0.440	**0.463**	0.034	0.355	0.370	0.369	0.382	0.372
30	0.448	**0.491**	0.122	0.359	0.391	0.373	0.388	0.374
35	0.450	**0.458**	0.182	0.376	0.389	0.384	0.395	0.370
40	**0.444**	0.407	0.244	0.375	0.385	0.396	0.393	0.375
45	**0.452**	0.439	0.313	0.379	0.388	0.395	0.396	0.374
50	**0.421**	0.413	0.318	0.380	0.405	0.399	0.390	0.370

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 1: Complete forgetting graph

Train edges: mastery 279,895; forgetting 1,167,562; recommendation 36,350 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.168	**0.188**	-0.007	0.150	0.094	0.144	0.187	0.155
10	0.213	**0.311**	-0.030	0.237	0.230	0.239	0.299	0.247
15	0.331	**0.458**	0.021	0.279	0.322	0.294	0.351	0.310
20	0.384	**0.454**	-0.017	0.315	0.329	0.324	0.391	0.363
25	0.430	**0.455**	-0.003	0.337	0.345	0.356	0.367	0.360
30	0.440	**0.485**	0.071	0.340	0.367	0.363	0.374	0.360
35	0.442	**0.455**	0.125	0.358	0.366	0.376	0.381	0.358
40	**0.437**	0.406	0.179	0.358	0.363	0.388	0.381	0.366
45	**0.445**	0.438	0.236	0.363	0.367	0.389	0.385	0.367
50	**0.415**	0.413	0.234	0.365	0.388	0.393	0.381	0.366

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.144	**0.167**	-0.015	0.130	0.060	0.126	0.158	0.131
10	0.168	**0.282**	-0.067	0.202	0.171	0.200	0.263	0.208
15	0.269	**0.421**	-0.027	0.237	0.253	0.243	0.313	0.265
20	0.309	**0.409**	-0.068	0.265	0.247	0.270	0.352	0.312
25	0.345	**0.408**	-0.056	0.281	0.254	0.299	0.325	0.308
30	0.347	**0.433**	0.008	0.280	0.276	0.302	0.333	0.307
35	0.347	**0.401**	0.053	0.293	0.273	0.312	0.342	0.304
40	0.339	**0.358**	0.100	0.290	0.269	0.325	0.340	0.309
45	0.345	**0.396**	0.158	0.291	0.275	0.323	0.342	0.311
50	0.314	**0.378**	0.146	0.293	0.291	0.326	0.336	0.305

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Remove zero-forgetting edges

Train edges: mastery 279,895; forgetting 694,068; recommendation 36,350 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	**0.880**	0.800	0.872	0.858	0.872	0.860	0.842	0.867
10	**0.879**	0.795	0.871	0.855	0.871	0.857	0.847	0.865
15	**0.876**	0.809	0.872	0.855	0.870	0.856	0.848	0.862
20	**0.874**	0.816	0.869	0.854	0.870	0.857	0.848	0.860
25	**0.873**	0.822	0.869	0.853	0.869	0.857	0.847	0.858
30	**0.872**	0.828	0.871	0.853	0.869	0.856	0.845	0.857
35	0.870	0.832	**0.871**	0.853	0.868	0.856	0.845	0.855
40	0.868	0.837	**0.872**	0.852	0.868	0.855	0.844	0.854
45	0.868	0.841	**0.871**	0.852	0.867	0.855	0.844	0.854
50	0.867	0.845	**0.871**	0.852	0.867	0.855	0.845	0.854

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	**1.000**	**1.000**	0.996	1.000	**1.000**	1.000	**1.000**	**1.000**
10	0.999	1.000	0.998	1.000	1.000	1.000	**1.000**	**1.000**
15	1.000	1.000	0.997	0.999	1.000	1.000	**1.000**	**1.000**
20	1.000	1.000	0.997	0.999	1.000	1.000	**1.000**	**1.000**
25	1.000	1.000	0.997	0.999	1.000	1.000	**1.000**	**1.000**
30	1.000	1.000	0.996	1.000	1.000	1.000	**1.000**	**1.000**
35	1.000	1.000	0.996	1.000	1.000	1.000	**1.000**	**1.000**
40	1.000	1.000	0.996	1.000	1.000	1.000	**1.000**	**1.000**
45	1.000	1.000	0.995	1.000	1.000	1.000	**1.000**	**1.000**
50	1.000	1.000	0.994	0.999	0.999	1.000	**1.000**	**1.000**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Remove zero-forgetting edges

Train edges: mastery 279,895; forgetting 694,068; recommendation 36,350 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.149	**0.260**	0.033	0.187	0.140	0.228	0.218	0.247
10	0.243	**0.382**	0.021	0.265	0.263	0.307	0.308	0.359
15	0.355	0.389	0.095	0.305	0.368	0.354	0.366	**0.399**
20	0.367	0.394	0.090	0.333	0.406	0.378	0.392	**0.419**
25	0.386	**0.458**	0.102	0.356	0.409	0.399	0.387	0.398
30	0.383	**0.481**	0.143	0.352	0.405	0.401	0.388	0.391
35	0.377	**0.569**	0.200	0.369	0.392	0.403	0.384	0.381
40	0.398	**0.565**	0.246	0.371	0.400	0.391	0.371	0.380
45	0.389	**0.486**	0.304	0.368	0.349	0.408	0.378	0.376
50	0.386	**0.488**	0.310	0.367	0.355	0.392	0.364	0.375

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.141	**0.255**	0.030	0.180	0.134	0.222	0.213	0.237
10	0.236	**0.379**	0.012	0.258	0.258	0.302	0.303	0.355
15	0.348	0.386	0.081	0.298	0.362	0.349	0.362	**0.397**
20	0.361	0.391	0.077	0.327	0.401	0.373	0.387	**0.417**
25	0.381	**0.457**	0.092	0.351	0.404	0.395	0.383	0.397
30	0.379	**0.480**	0.135	0.347	0.400	0.398	0.385	0.390
35	0.374	**0.569**	0.182	0.365	0.389	0.401	0.382	0.381
40	0.395	**0.564**	0.230	0.368	0.396	0.390	0.370	0.379
45	0.387	**0.486**	0.288	0.365	0.345	0.407	0.377	0.376
50	0.385	**0.488**	0.293	0.363	0.351	0.392	0.363	0.375

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Remove zero-forgetting edges

Train edges: mastery 279,895; forgetting 694,068; recommendation 36,350 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.119	**0.247**	0.027	0.170	0.114	0.214	0.207	0.223
10	0.212	**0.371**	-0.011	0.247	0.234	0.293	0.298	0.343
15	0.320	0.381	0.053	0.285	0.332	0.340	0.357	**0.386**
20	0.332	0.385	0.047	0.313	0.371	0.365	0.381	**0.408**
25	0.352	**0.452**	0.059	0.337	0.372	0.386	0.377	0.387
30	0.347	**0.478**	0.093	0.334	0.367	0.390	0.379	0.381
35	0.341	**0.568**	0.135	0.351	0.355	0.393	0.376	0.372
40	0.360	**0.564**	0.172	0.352	0.362	0.384	0.364	0.371
45	0.351	**0.486**	0.225	0.349	0.311	0.402	0.370	0.368
50	0.348	**0.488**	0.227	0.348	0.317	0.388	0.358	0.367

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.073	**0.225**	0.020	0.153	0.066	0.192	0.183	0.201
10	0.138	**0.348**	-0.039	0.223	0.164	0.264	0.275	0.318
15	0.228	**0.361**	0.011	0.261	0.243	0.307	0.329	0.361
20	0.231	0.362	0.006	0.282	0.271	0.327	0.352	**0.385**
25	0.245	**0.429**	0.009	0.304	0.266	0.344	0.346	0.367
30	0.234	**0.460**	0.035	0.298	0.261	0.347	0.348	0.361
35	0.223	**0.553**	0.068	0.311	0.243	0.347	0.343	0.353
40	0.240	**0.555**	0.096	0.317	0.252	0.337	0.330	0.351
45	0.233	**0.479**	0.140	0.313	0.199	0.353	0.334	0.350
50	0.231	**0.483**	0.135	0.308	0.203	0.340	0.320	0.350

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Weighted nonzero sampling, up to 400/student

Train edges: mastery 279,895; forgetting 290,800; recommendation 36,350 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.879	0.809	**0.880**	0.858	0.862	0.861	0.848	0.879
10	**0.877**	0.811	0.869	0.859	0.866	0.859	0.855	0.876
15	**0.874**	0.826	0.872	0.859	0.867	0.860	0.858	0.873
20	**0.873**	0.832	0.870	0.858	0.866	0.861	0.858	0.869
25	0.871	0.836	**0.871**	0.857	0.866	0.860	0.856	0.866
30	0.870	0.839	**0.872**	0.857	0.866	0.860	0.854	0.864
35	0.869	0.842	**0.872**	0.856	0.866	0.860	0.853	0.863
40	0.869	0.846	**0.873**	0.856	0.866	0.859	0.852	0.862
45	0.868	0.849	**0.873**	0.856	0.866	0.859	0.852	0.860
50	0.868	0.852	**0.874**	0.856	0.865	0.858	0.851	0.859

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	**1.000**	1.000	0.997	1.000	**1.000**	**1.000**	**1.000**	**1.000**
10	**1.000**	1.000	0.996	1.000	1.000	1.000	**1.000**	**1.000**
15	**1.000**	1.000	0.995	1.000	1.000	1.000	**1.000**	**1.000**
20	**1.000**	1.000	0.996	1.000	1.000	1.000	**1.000**	**1.000**
25	**1.000**	1.000	0.996	1.000	1.000	1.000	**1.000**	**1.000**
30	**1.000**	1.000	0.995	1.000	1.000	1.000	**1.000**	**1.000**
35	**1.000**	1.000	0.994	1.000	1.000	1.000	**1.000**	**1.000**
40	1.000	1.000	0.994	1.000	1.000	1.000	**1.000**	**1.000**
45	1.000	1.000	0.993	0.999	1.000	1.000	**1.000**	**1.000**
50	1.000	1.000	0.993	1.000	1.000	1.000	**1.000**	**1.000**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Weighted nonzero sampling, up to 400/student

Train edges: mastery 279,895; forgetting 290,800; recommendation 36,350 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.171	0.212	0.062	0.179	0.178	0.192	0.236	**0.247**
10	0.244	0.297	0.092	0.259	0.304	0.254	0.308	**0.320**
15	0.319	0.305	0.030	0.308	**0.390**	0.318	0.368	0.368
20	0.397	0.334	0.033	0.336	0.382	0.344	**0.404**	0.391
25	**0.405**	0.386	0.073	0.346	0.389	0.353	0.387	0.401
30	0.400	**0.445**	0.119	0.348	0.383	0.360	0.385	0.405
35	0.392	**0.460**	0.197	0.356	0.376	0.369	0.375	0.399
40	0.395	**0.421**	0.231	0.362	0.373	0.368	0.374	0.393
45	0.416	**0.429**	0.247	0.367	0.388	0.378	0.371	0.386
50	0.427	**0.435**	0.304	0.368	0.372	0.378	0.366	0.378

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.165	0.208	0.062	0.176	0.172	0.187	0.233	**0.239**
10	0.240	0.293	0.080	0.255	0.299	0.249	0.305	**0.314**
15	0.317	0.303	0.018	0.304	**0.387**	0.313	0.365	0.363
20	0.396	0.332	0.022	0.330	0.381	0.340	**0.401**	0.387
25	**0.404**	0.384	0.058	0.341	0.388	0.349	0.384	0.399
30	0.400	**0.444**	0.106	0.344	0.383	0.356	0.382	0.403
35	0.391	**0.459**	0.178	0.353	0.376	0.367	0.372	0.398
40	0.395	**0.421**	0.216	0.360	0.373	0.366	0.372	0.392
45	0.416	**0.429**	0.228	0.366	0.387	0.376	0.370	0.386
50	0.427	**0.435**	0.286	0.367	0.368	0.377	0.365	0.378

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Weighted nonzero sampling, up to 400/student

Train edges: mastery 279,895; forgetting 290,800; recommendation 36,350 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.151	0.198	0.062	0.172	0.150	0.180	**0.227**	0.225
10	0.221	0.284	0.055	0.249	0.278	0.238	**0.296**	0.294
15	0.294	0.295	-0.010	0.297	**0.366**	0.300	0.357	0.339
20	0.377	0.325	-0.008	0.324	0.365	0.325	**0.395**	0.365
25	**0.386**	0.377	0.028	0.333	0.375	0.333	0.379	0.380
30	0.383	**0.438**	0.067	0.335	0.372	0.339	0.377	0.388
35	0.376	**0.456**	0.138	0.342	0.366	0.349	0.367	0.384
40	0.382	**0.419**	0.172	0.348	0.362	0.348	0.367	0.381
45	0.403	**0.428**	0.174	0.354	0.379	0.359	0.365	0.375
50	0.412	**0.435**	0.225	0.356	0.357	0.363	0.362	0.369

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.120	0.180	0.058	0.150	0.120	0.156	**0.208**	0.189
10	0.167	0.258	0.004	0.218	0.226	0.197	**0.279**	0.251
15	0.218	0.269	-0.074	0.259	0.300	0.254	**0.340**	0.294
20	0.293	0.300	-0.075	0.281	0.294	0.274	**0.372**	0.317
25	0.299	0.352	-0.051	0.282	0.295	0.280	**0.357**	0.324
30	0.295	**0.409**	-0.034	0.276	0.297	0.283	0.354	0.331
35	0.290	**0.427**	0.021	0.283	0.290	0.294	0.347	0.324
40	0.295	**0.396**	0.037	0.288	0.276	0.292	0.341	0.314
45	0.317	**0.410**	0.037	0.295	0.296	0.300	0.343	0.310
50	0.329	**0.418**	0.080	0.297	0.275	0.303	0.339	0.302

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 4: Weighted nonzero sampling, up to 200/student

Train edges: mastery 279,895; forgetting 145,400; recommendation 36,350 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.879	0.803	0.882	0.862	0.869	0.862	0.870	**0.889**
10	0.877	0.824	0.880	0.860	0.868	0.860	0.873	**0.886**
15	0.876	0.828	0.876	0.859	0.868	0.860	0.872	**0.881**
20	0.874	0.831	0.874	0.859	0.869	0.860	0.870	**0.877**
25	0.872	0.833	0.871	0.858	0.868	0.859	0.866	**0.874**
30	0.871	0.837	0.871	0.857	0.867	0.858	0.862	**0.871**
35	0.869	0.841	**0.871**	0.857	0.867	0.857	0.860	0.869
40	0.869	0.845	**0.872**	0.856	0.866	0.858	0.858	0.868
45	0.868	0.847	**0.871**	0.856	0.866	0.857	0.857	0.866
50	0.868	0.849	**0.871**	0.856	0.866	0.857	0.856	0.864

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	**1.000**	**1.000**	0.997	1.000	0.998	1.000	**1.000**	**1.000**
10	**1.000**	1.000	0.998	1.000	0.999	1.000	**1.000**	**1.000**
15	**1.000**	1.000	0.998	1.000	0.999	1.000	**1.000**	**1.000**
20	**1.000**	1.000	0.997	1.000	0.999	1.000	**1.000**	**1.000**
25	**1.000**	1.000	0.996	1.000	0.999	1.000	**1.000**	**1.000**
30	**1.000**	1.000	0.996	1.000	0.999	1.000	**1.000**	**1.000**
35	1.000	1.000	0.996	1.000	0.999	1.000	**1.000**	**1.000**
40	1.000	1.000	0.995	1.000	0.999	1.000	**1.000**	**1.000**
45	1.000	1.000	0.993	1.000	0.999	1.000	**1.000**	**1.000**
50	1.000	1.000	0.992	1.000	0.999	1.000	**1.000**	**1.000**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 4: Weighted nonzero sampling, up to 200/student

Train edges: mastery 279,895; forgetting 145,400; recommendation 36,350 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.165	0.245	0.045	0.184	0.126	0.179	0.218	**0.247**
10	0.247	0.278	0.036	0.274	0.234	0.261	0.306	**0.313**
15	0.294	0.309	0.048	0.310	0.239	0.319	**0.356**	0.355
20	0.308	0.370	0.035	0.341	0.334	0.349	**0.397**	0.374
25	0.349	**0.399**	0.063	0.347	0.368	0.347	0.391	0.372
30	0.371	**0.444**	0.121	0.349	0.340	0.371	0.385	0.381
35	0.351	**0.452**	0.217	0.364	0.368	0.373	0.372	0.367
40	0.379	**0.459**	0.279	0.365	0.394	0.366	0.375	0.357
45	0.384	**0.425**	0.332	0.382	0.408	0.362	0.374	0.358
50	0.382	**0.401**	0.354	0.376	0.391	0.358	0.372	0.366

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.163	**0.240**	0.042	0.181	0.117	0.175	0.210	0.240
10	0.245	0.273	0.026	0.270	0.226	0.257	0.302	**0.308**
15	0.291	0.304	0.038	0.305	0.231	0.316	**0.352**	0.350
20	0.305	0.366	0.025	0.336	0.327	0.347	**0.393**	0.371
25	0.346	**0.395**	0.051	0.342	0.363	0.345	0.387	0.370
30	0.368	**0.441**	0.111	0.344	0.335	0.370	0.382	0.379
35	0.349	**0.451**	0.200	0.360	0.365	0.372	0.369	0.366
40	0.376	**0.459**	0.264	0.362	0.391	0.366	0.373	0.357
45	0.382	**0.425**	0.318	0.380	0.405	0.362	0.373	0.358
50	0.381	**0.401**	0.340	0.374	0.389	0.357	0.372	0.366

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 4: Weighted nonzero sampling, up to 200/student

Train edges: mastery 279,895; forgetting 145,400; recommendation 36,350 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.152	**0.235**	0.038	0.174	0.083	0.166	0.203	0.232
10	0.232	0.267	0.004	0.261	0.191	0.244	0.294	**0.302**
15	0.277	0.299	0.012	0.297	0.198	0.302	0.344	**0.344**
20	0.290	0.362	-0.005	0.326	0.291	0.333	**0.385**	0.366
25	0.331	**0.391**	0.018	0.331	0.328	0.331	0.380	0.364
30	0.353	**0.436**	0.069	0.335	0.296	0.355	0.375	0.373
35	0.334	**0.448**	0.143	0.351	0.325	0.359	0.363	0.360
40	0.359	**0.458**	0.206	0.353	0.352	0.353	0.367	0.351
45	0.369	**0.425**	0.248	0.371	0.366	0.349	0.368	0.353
50	0.368	**0.401**	0.265	0.365	0.349	0.344	0.367	0.362

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.114	**0.216**	0.033	0.156	0.024	0.145	0.176	0.208
10	0.161	0.240	-0.028	0.232	0.120	0.211	0.260	**0.274**
15	0.202	0.264	-0.037	0.264	0.120	0.261	0.303	**0.311**
20	0.206	0.321	-0.055	0.285	0.203	0.280	**0.342**	0.324
25	0.244	**0.347**	-0.041	0.289	0.238	0.271	0.334	0.324
30	0.264	**0.389**	0.005	0.286	0.190	0.291	0.325	0.329
35	0.237	**0.399**	0.061	0.301	0.228	0.295	0.312	0.313
40	0.262	**0.409**	0.106	0.295	0.249	0.287	0.314	0.302
45	0.264	**0.384**	0.142	0.316	0.266	0.283	0.318	0.301
50	0.273	**0.365**	0.147	0.308	0.240	0.279	0.317	0.308

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.